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SPECIAL REPORT

Disease-modifying anti-nephropathic drugs (DMANDs)—a definition proposed by the Immunonephrology Working Group (IWG) of the European Renal Association (ERA)

Y.K. Onno Teng [1,*], Eleni Frangou [2,3,4,*], Andreas Kronbichler [5], Annette Bruchfeld [6,7], Fernando Caravaca-Fontan [8], Jürgen Floege [9], Sarah M. Moran [10], Safak Mirioglu [11], Kate I. Stevens [12], Stefanie Steiger [13], Paola Romagnani [14,15] and Hans-Joachim Anders [13]; on behalf of the Immunonephrology Working Group of the European Renal Association

[1]Center of Expertise for Lupus-, Vasculitis- and Complement-Mediated Systemic Diseases (LuVaCs), Department of Internal Medicine—Nephrology Section, Leiden University Medical Center, Leiden, The Netherlands
[2]Department of Nephrology, Limassol General Hospital, State Health Services Organization, Limassol, Cyprus
[3]University of Nicosia Medical School, Nicosia, Cyprus
[4]Medical School, National and Kapodistrian University of Athens, Athens, Greece
[5]Department of Internal Medicine IV, Nephrology and Hypertension, Medical University Innsbruck, Innsbruck, Austria
[6]Department of Health, Medicine and Caring Sciences, Linköping University, Linköping, Sweden
[7]Department of Renal Medicine, Karolinska University Hospital and CLINTEC Karolinska Institutet, Stockholm, Sweden
[8]Department of Nephrology, Instituto de Investigación Hospital 12 de Octubre (imas12), Madrid, Spain
[9]Division of Nephrology, RWTH Aachen University Hospital, Aachen, Germany
[10]Department of Nephrology, Cork University Hospital & School of Medicine, University College Cork, Cork, Ireland
[11]Department of Immunology, Aziz Sancar Institute of Experimental Medicine, Istanbul University, Istanbul, Turkey
[12]Glasgow Renal and Transplant Unit, Queen Elizabeth University Hospital, Glasgow, UK
[13]Division of Nephrology, Department of Internal Medicine IV, Hospital of the Ludwig-Maximilians-University, Munich, Germany
[14]Nephrology and Dialysis Unit, Meyer Children's Hospital IRCCS, Florence, Italy
[15]Department of Biomedical, Experimental and Clinical Sciences "Mario Serio", University of Florence, Florence, Italy
Correspondence to: Hans-Joachim Anders; E-mail: hjanders@med.uni-muenchen.de
*Equal contribution.

ABSTRACT

A definition of 'disease modification' for kidney disease is long overdue. Here, we propose three key criteria for disease modification in immune-mediated glomerulonephritis and podocytopathies: minimizing disease activity, preventing loss of kidney structure and function, and reducing treatment-related toxicity. To be considered a disease-modifying anti-nephropathic drug (DMAND), a drug must fulfil all three criteria, hence the DMAND status of a drug may not be clear at the time of regulatory approval. Notably, the aspect of chronic kidney disease (CKD) in immune-mediated kidney diseases must be considered and treated separately, e.g. renin–angiotensin system inhibitor is a DMAND for the CKD aspect but not for the immune disease itself. Defining DMANDs is an ambitious goal but one that may help to set the priorities for future treatment strategies in immune-mediated kidney disease. This may mean much more rapid tapering or even avoidance of unselective, non-targeted immunosuppressive agents, which carry considerable short (teratogenicity) and long term risks (malignancies). The criteria proposed here set a high bar for 'disease modification' in immune-mediated kidney disease. Inevitably, this must dictate altered priorities with the focus for new therapeutic agents and strategies shifting from solely reduction of proteinuria to preservation of GFR and attenuation of decline, whilst also eliminating long-term toxicity.

Keywords: disease modification, disease modifying drug, glomerulonephritis, immune-mediated kidney disease, immunonephrology

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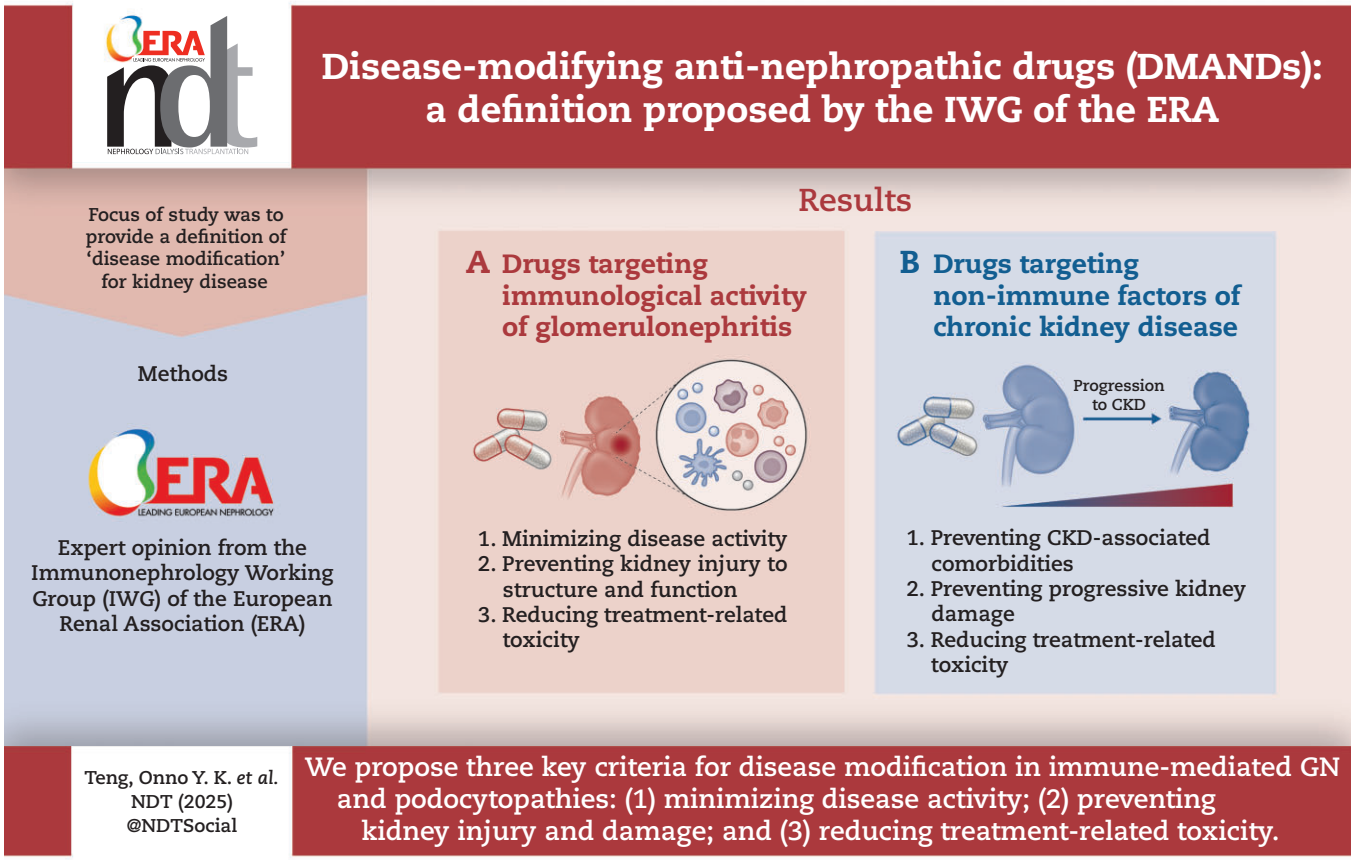
GRAPHICAL ABSTRACT



Disease modification is generally defined as 'an improvement of symptoms in conjunction with the change of the disease course', i.e. disease outcome [1]. Symptoms and outcomes differ from one disease to another thus, disease modification has to be specified for each disorder separately. Every branch of medicine faces unique challenges in defining 'disease modification' because symptoms may be difficult to dissect from irreversible functional impairment, as is the case in Parkinson's disease [2], and because disease outcomes can be diverse. For example, in inflammatory bowel disease, outcomes might include disability, bowel surgery, stoma or cancer [2, 3].

The concept of 'disease modification' is easier to embrace in diseases where symptom-driven treatment is possible. In rheumatoid arthritis, arthralgia and joint stiffness are symptoms controlled with steroids and non-steroidal anti-inflammatory drugs (NSAIDs); however, disease modification in this disease also requires protection against joint destruction and prevention of functional disability, which is not offered by these drugs [1]. Thus in rheumatoid arthritis, steroids and NSAIDs are not disease-modifying but only symptomatic treatments. In contrast, methotrexate, tumour necrosis factor blockers and other immunomodulators interfere with the immunopathogenesis of the disease and hence attenuate joint destruction and consequent functional disability. It is this that defines them as 'disease-modifying anti-rheumatic drugs' (DMARDs). Today, whilst steroids and NSAIDs remain pivotal for symptom control, the DMARD concept has been instrumental in defining priorities for drug developers and clinicians in rheumatoid arthritis but also in other chronic diseases that may have significant individual and societal level impact in terms of disability, quality of life and socioeconomic consequences.

Currently, the field of immune-mediated kidney disease is experiencing a rapid increase in the therapeutic options available for autoimmune glomerulonephritis, vasculitis and podocytopathies. This expanding therapeutic armamentarium prompts nephrologists to address the concept of 'disease modification' for immune-mediated kidney disease to guide drug use and development as well as clinical decision-making on outcome priorities. Recently, criteria for defining drugs with the potential for disease modification have been proposed for systemic lupus erythematosus [4], a systemic disorder leading to immune-complex glomerulonephritis (lupus nephritis). These include 'minimizing disease activity with the fewest treatment-associated toxicities and slowing or preventing organ damage progression' [4]. This approach could be useful to dichotomize 'disease-modifying anti-nephropathic drugs' (DMANDs) from drugs that only provide control of symptoms or functional abnormalities. For lupus nephritis, DMANDs could be defined as drugs that fulfil three citeria: minimize disease activity, prevent progressive kidney injury and damage, and avoid treatment-related toxicity. Similar criteria would apply to identify DMANDs used for other forms of autoimmune glomerulonephritis, including anti-neutrophil antibody–associated vasculitis, immunoglobulin A (IgA) nephropathy, anti-glomerular basement membrane disease and primary membranous nephropathy (Supplementary data, Table S1).

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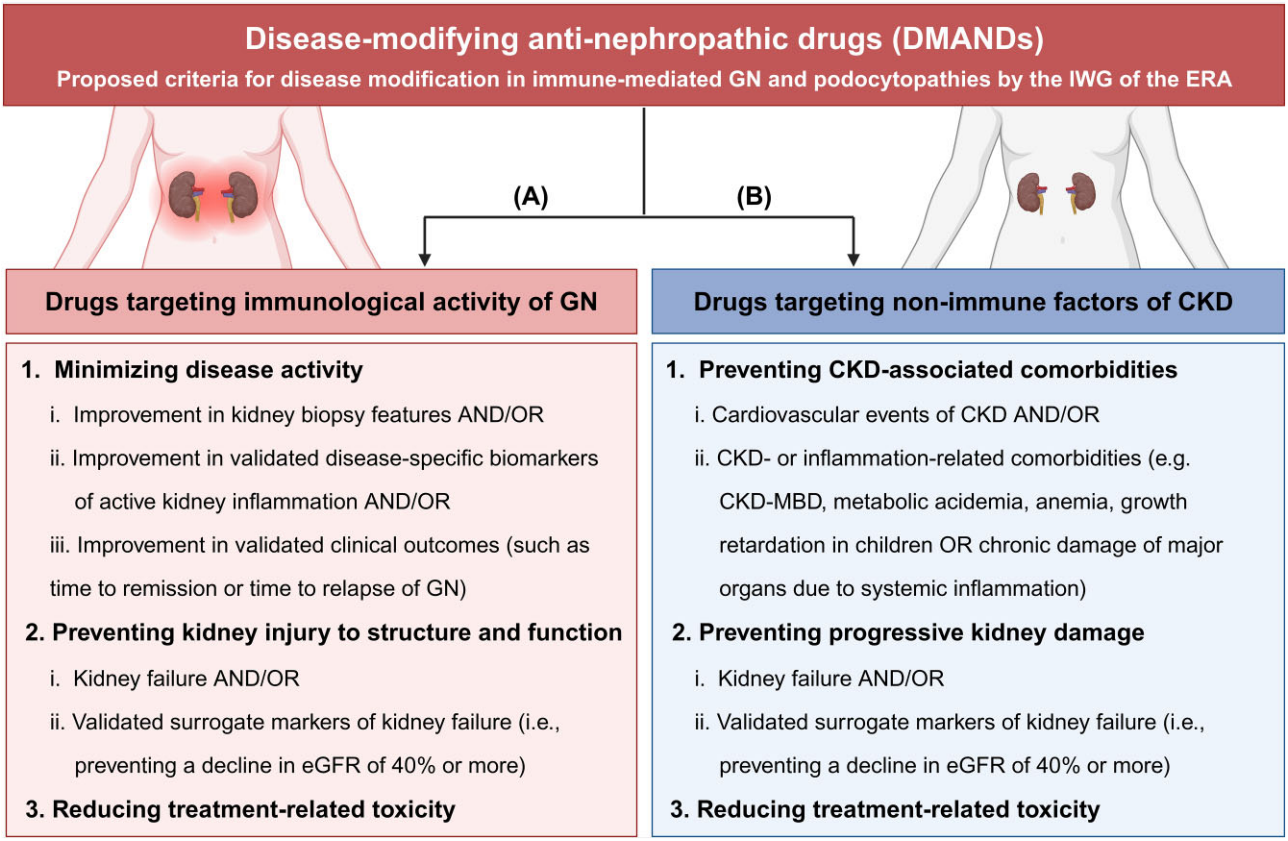

Figure 1: European Renal Association (ERA) criteria for DMANDs for glomerulonephritis and immune-mediated podocytopathies. (**A**) Criteria for drugs targeting the immunological activity of glomerulonephritis. Drug must meet all three criteria. (**B**) Criteria for drugs targeting the non-immune factors of CKD. Drug must meet both criteria. ISN/RPS, International Society of Nephrology/Renal Pathology Society; GN, glomerulonephritis.

Defining DMANDs is not without controversy because robust data on long-term outcomes are not yet available for many novel drugs with a mechanism-of-action that interferes with the immunopathogenesis of disease. Whilst many of these new drugs, such as targeted-release formulation budenoside in IgA nephropathy or CD38 blockade in membranous nephropathy or lupus nephritis, have demonstrated positive short-term effects (i.e. proteinuria response), it remains uncertain whether these will translate into long-term benefits (i.e. reduced risk of kidney failure and fewer adverse events) that reach beyond surrogate markers of kidney failure. Therefore, the proposed definition of DMANDs (Fig. 1) intends to raise the awareness of the nephrology field and highlight the need for new therapeutic strategies that aim for disease modification.

Importantly, DMAND criteria are not necessarily identical to primary endpoints of clinical trials or validated (surrogate) endpoints used by regulatory bodies for drug approval or reimbursement criteria by political entities or medical insurances. A drug with anti-proteinuric effects but without evidence of a protective effect on the kidney or one with treatment-associated toxicity would not meet the proposed DMAND criteria. DMAND criteria are deliberately more rigid because they focus on the long-term impact on the disease process, which is rarely fully captured in clinical trials and usually necessitates long term evaluation. Nevertheless, the DMAND criteria proposed here are in line with the criteria used by the European Medicines Agency for the assessment of medicinal products slowing the progression of chronic kidney disease (CKD) [5].

The same is also applicable for immune-mediated podocytopathies, which traditionally rely on a proteinuria response as a primary endpoint in clinical trials. The incidence and prevalence of immune-mediated podocytopathies is lower than other kidney disease and so undertaking clinical trials with CKD endpoints is more challenging in this scenario. However, as evidenced by the DUPLEX-FSGS trial, global trial networks can overcome this hurdle. In this trial, proteinuria response did not translate into a significant reduction of estimated glomerular filtration rate (eGFR) loss during the duration of the study [6]. Moreover, the MENTOR trial revealed in primary membranous nephropathy that the anti-proteinuric effect of cyclosporine A comes at the expense of a more rapid GFR decline, an important argument to insist on DMAND criteria when assessing immune-mediated podocytopathies [7]. A further example in the podocytopathy field is the apolipoprotein (APO) L1 inhibitor, inaxaplin, which demonstrated an anti-proteinuric effect in patients with genetic APOL1 risk constellations [8]. Whether it can also prevent kidney function decline is currently under study (NCT05312879).

Could a similar DMAND definition be applied to genetic nephropathies, interstitial nephritis or thrombotic microangiopathies? If the focus of the definition is on disease outcomes, a DMAND definition of these outcomes should address disease modification of the underpinning pathophysiology of each disease as suggested in Supplementary data, Table S1. However, it will be crucial to define what signifies 'minimizing disease activity with the fewest treatment-associated toxicities and slowing or preventing organ damage progression' for each of these

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diseases separately. Terms such as 'disease activity', 'proteinuria' and 'repeat biopsy' cannot be applied to cystic or other genetic kidney diseases or interstitial nephritis. For thrombotic microangiopathies, which follow a different dynamic, additional biomarkers like platelet count, haptoglobulin or even complement levels could be utilized to assess activity.

In addition, an important consideration is the CKD aspect of each of these disorders. The primary goal of DMANDs is to reduce the decline in eGFR and prevent kidney failure; yet, in many of these conditions a degree of CKD is inevitable. Therapies like inhibitors of the renin–angiotensin system (RASi) and of the sodium-glucose cotransporter 2 (SGLT2i) have no apparent effect on the disease activity of immune-mediated kidney disease, but they do reduce proteinuria and improve long-term kidney and cardiovascular outcomes in diabetic and non-diabetic CKD. To be defined as a DMAND, it is mandated that a drug controls disease activity by exhibiting a mechanism-of-action that interferes with the molecular mechanisms of kidney injury and inflammation. Therefore, RASi and SGLT2i are not classified as DMANDs from the perspective of immune-mediated kidney disease (Fig. 1A) but from the perspective of CKD (Fig. 1B). Therefore, kidney activity and chronicity are recognized as additive risks and treated separately in immune-mediated kidney disease, hence criteria for 'disease modification' differ for these two aspects of glomerulonephritis and immune-mediated podocytopathies [9, 10].

In summary, a definition of 'disease modification' for kidney disease is long overdue. Here, we propose three key criteria for disease modification in immune-mediated glomerulonephritis and podocytopathies: minimizing disease activity, preventing kidney injury to structure and function, and reducing treatment-related toxicity (Fig. 1). In principle, a drug should fulfil all three criteria to be considered a DMAND. The criteria for DMANDs aim to raise the awareness and highlight the need of the nephrology community to address and aim for disease modification in immune-mediated kidney diseases. It is self-evident that the DMAND status of a drug may not be clear at the time of regulatory approval which is predominantly based on superior clinical efficacy. Notably, the aspect of CKD in immune-mediated kidney diseases must be considered and treated separately, e.g. RASi is a DMAND for the CKD aspect but not for the immune disease itself. Defining DMANDs is an ambitious goal but one that may help to set the priorities for future treatment strategies in immune-mediated kidney diseases. This may mean much more rapid tapering or even avoidance of unselective, non-targeted immunosuppressive agents, which carry considerable short- (teratogenicity) and long-term (malignancies) risks. The criteria proposed here set a high bar for 'disease modification' in immune-mediated kidney disease. Inevitably, this must dictate altered priorities with the focus for new therapeutic agents and strategies shifting from solely reduction of proteinuria to preservation of GFR and attenuation of decline, whilst also eliminating long-term toxicity.

SUPPLEMENTARY DATA

Supplementary data are available at *Nephrology Dialysis Transplantation* online.

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AUTHORS' CONTRIBUTIONS

H.-J.A., Y.K.O.T. and E.F. drafted the first version of the article. All authors significantly revised the content and reviewed the final version of the manuscript.

DATA AVAILABILITY STATEMENT

No new data were generated or analysed in support of this research.

CONFLICT OF INTEREST STATEMENT

Outside the submitted work, A.K. received support for meeting registration and travel from AstraZeneca and Otsuka, an unrestricted research grant from CSL Vifor, and consultancy fees or speaking fees from AstraZeneca, Catalyst Biosciences, CSL Vifor, Delta, GlaxoSmithKline, Miltenyi Biotech, Novartis, Roche and Walden Biosciences. H.-J.A. received consultancy or lecture fees from Boehringer-Ingelheim, Bayer, GSK, AstraZeneca, Novartis, Otsuka, Roche, SOBI and Janssen, and serves as Editor-in-Chief of *NDT*. A.B. received consultancy fees from Boehringer-Ingelheim and CSL-Vifor, and payment or honoraria for lectures, presentations, speaker's bureaus, manuscript writing or educational events from AstraZeneca, Bayer, Boehringer-Ingelheim, Chemocentryx, CSL-Vifor, Fresenius, GSK and Otsuka. F.C.-F. received consultancy fees from Novartis, SOBI and Apellis, payment or honoraria for lectures, presentations, speaker's bureaus, manuscript writing or educational events from AstraZeneca, and participated on a data safety monitoring board or advisory board for Novartis and SOBI. J.F. received consultancy fees from AstraZeneca, Bayer, Boehringer, Calliditas, CSL Vifor, HiBio, Novartis, Omeros, Roche, Travere and Vera Tx, payment or honoraria for lectures, presentations, speaker's bureaus, manuscript writing or educational events from AstraZeneca, Bayer, Boehringer, Calliditas, CSL Vifor, HiBio, Novartis, Omeros, Roche, Stadapharm, Travere and Vera Tx, and participated on a data safety monitoring board or advisory board for NovoNordisk. E.F. received grants or contracts, and support for meeting registration and travel from State Health Services Organization of Cyprus. K.I.S. received consultancy fees from CSL Vifor, Boehringer Ingelheim and Bayer. S.M. received support for meeting registration and travel from Amgen and Sanofi Genzyme. S.M.M. received support for meeting registration and travel from CSL Vifor and Boehringer Ingelheim, an unrestricted research grant from CSL Vifor, and consultancy fees or speaking fees from AstraZeneca, CSL Vifor and Boehringer Ingelheim. E.F., A.K., S.M. and S.S. are Editorial Board members of *NDT*. The remaining authors have no disclosures.

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